

IAL Biology Unit 5 (WBI15): Rapid Revision

Respiration, Internal Environment, Coordination and Gene Technology · last-minute summary · 1 h 45 min · 90 marks · Q1-Q7 = 70 marks · the final question is 20 marks on the pre-released article · budget 25 minutes for it

WHERE THE MARKS ARE

Nerves and synapses (10) · heart and exercise (10) · kidney (8) · respiration (8) · gene expression (7) · muscle (7). **Nerves, heart, kidney and respiration carry about 36 of the 70 non-article marks.**

WHAT THIS PAPER REWARDS

1. Exact wording — the schemes reject “signals”, “produce energy”, “ATPase” and “sodium” unqualified. **2.** Chains in **the right order**: several schemes say “mark points need to be in logical order”. **3.** Name which groups when commenting on error bars. **4.** Add a methodology comment — often the easiest mark. **5.** Level 3 needs the **molecular** mechanism, not the physiology.

RESPIRATION

Stage	Where	Products
Glycolysis	Cytoplasm	2 pyruvate, net 2 ATP , 2 reduced NAD
Link reaction	Matrix	Pyruvate oxidised and decarboxylated → acetyl + coenzyme A ; CO ₂ ; reduced NAD
Krebs cycle	Matrix	Per turn: 2 CO₂ , reduced NAD, reduced FAD, 1 ATP
Oxidative phosph.	Inner membrane / cristae	Most of the ATP; water

CHEMIOSMOSIS — THE MOST-ASKED CHAIN IN UNIT 5

1. Reduced NAD and FAD carry hydrogen to the electron transport chain on the **inner membrane (cristae)**. **2.** Hydrogen **splits into protons and electrons**. **3.** Electrons pass along the chain, **releasing energy** — “**produce energy**” is rejected. **4.** Energy **pumps protons from the matrix into the intermembrane space**. **5.** Protons diffuse back through **ATP synthase** — chemiosmosis. **6.** ADP + P_i → ATP. **7. Oxygen is the final electron acceptor**, forming water.

Reject “ATPase” — write ATP synthase. “ADP + P_i → ATP without reference to ATP synthase” is ignored. Location: **inner membrane or cristae** — “membrane” or “intermembrane space” is ignored.

Inhibitor logic is always the same: block the chain → oxygen not used → no proton gradient → no oxidative phosphorylation → only glycolysis → lactate → pH falls → enzymes denature.

Anaerobic: the chain stops → reduced NAD is not oxidised → pyruvate is **reduced to lactate, regenerating NAD** so glycolysis continues (2 ATP). **Lactate goes to the liver and is oxidised back to pyruvate** — it is **never** “used in the Krebs cycle” directly.

Oxygen uptake stays high after exercise because oxygen

is needed to convert lactate back to pyruvate or glucose, replenish glycogen, re-oxygenate myoglobin and repay the oxygen debt.

RQ = volume of CO₂ produced ÷ volume of O₂ consumed in the same time. Both the ratio **and** “volume produced / volume used” are needed; “amount” is ignored. Carbohydrate **1.0**, protein **~0.9**, lipid **0.7**, anaerobic carbohydrate **above 1.0**. **No units.**

MUSCLE

SLIDING FILAMENT — POINTS MUST BE IN LOGICAL ORDER

1. Action potential along the sarcolemma and down the **T-tubules**. **2. Ca²⁺ released from the sarcoplasmic reticulum**. **3. Ca²⁺ binds troponin**, which changes shape. **4.** Troponin moves **tropomyosin**, exposing the **myosin-binding sites on actin**. **5.** Myosin heads bind → **cross-bridges**. **6. ADP and P_i released** → head tilts — the **power stroke** — pulling actin towards the centre. **7. ATP binds**, breaking the cross-bridge. **8. ATP hydrolysed**; energy returns the head to its original position. **9.** Repeats while Ca²⁺ is present; Ca²⁺ is actively transported back to relax.

The repeated trap: ATP hydrolysis moves the head **back** — **do not call that the power stroke**. The power stroke is driven by the **release of ADP and P_i**.

Band	Contains	On contraction
Z line	Ends of sarcomere	Move closer
I band	Actin only	Shortens
A band	Full length of myosin	Unchanged
H zone	Myosin only	Shortens

Feature	Slow twitch	Fast twitch
Respiration	Aerobic	Anaerobic
Mitochondria, myoglobin, capillaries	Many — red	Few — pale
Sarcoplasmic reticulum	Less	Extensive
Contraction	Slow, sustained	Rapid, powerful, fatigues

Ligament joins bone to bone, elastic, **maintains joint stability**. **Tendon** joins muscle to bone, **inelastic, transmits force to the bones**. Only **cardiac** muscle is myogenic.

HEART, VENTILATION AND EXERCISE

Myogenic = contraction is initiated **within the cardiac muscle itself, without a nerve impulse**.

THE CONDUCTION CHAIN

SAN in the right atrium wall is the pacemaker, generating **waves of depolarisation or excitation** — “**impulses**” is **ignored** here → spreads across both atria → **atrial systole** → **non-conducting tissue** stops it reaching the ventricles directly → the **AVN delays** the wave so the atria finish emptying → **bundle of His** in the septum → **Purkyne fibres** → ventricles contract **from the apex upwards**.

ECG: P = atrial depolarisation; QRS = ventricular depolarisation; T = ventricular **repolarisation**. When comparing traces you **must name the PQRST waves** or the marks are not awarded. Heart rate = $60 \div \text{time for one cycle}$; cardiac output = heart rate $\times$ stroke volume.

CONTROL OF HEART RATE

Exercise → more respiration → **more CO₂, so blood pH falls** → **chemoreceptors** in the carotid and aortic bodies detect it → impulses to the **cardiovascular control centre in the medulla** → more impulses along **sympathetic (accelerator) neurones to the SAN** → the SAN **depolarises more frequently** → rate and force of contraction increase. To slow it: the **parasympathetic or vagus nerve**.

Do not accept “signals” or “messages”. “Increased oxygen demand” as the stimulus is ignored — receptors detect **CO₂ and pH**.

Ventilation: at rest, **stretch receptors** in the lungs inhibit the inspiratory centre. In exercise, more CO₂ and lower pH → chemoreceptors → respiratory centre in the medulla → more impulses to the **intercostal muscles and diaphragm** → rate and depth increase.

Blood distribution in exercise: cardiac output up; flow to skeletal and cardiac muscle up; to the **skin up** (thermoregulation); to kidneys and gut **down**; to the **brain unchanged**.

Trained athletes: larger, stronger cardiac muscle → greater stroke volume → **lower resting heart rate**; more slow-twitch fibres, mitochondria, myoglobin and capillaries; they switch to anaerobic respiration later, so less lactate.

HOMEOSTASIS, HORMONES AND THE KIDNEY

Homeostasis = maintaining the body in a state of **dynamic equilibrium**. **Negative feedback reverses** a change; **positive feedback amplifies** it. The point most answers omit: **when receptors detect the condition is back to normal, the effector stops being stimulated**.

Thermoregulation: **thermoreceptors** in the skin and hypothalamus → **hypothalamus** → effectors. Too hot: **vasodilation of skin arterioles**, sweating, erector muscles relax. Too cold: vasoconstriction, shivering. “**Control centre**” **unqualified is ignored** — name the hypothalamus.

	Peptide	Steroid
Solubility	Water-soluble; cannot cross the membrane	Lipid-soluble; diffuses through
Receptor	On the cell surface	Inside the cell
Then	Activates a second messenger (cAMP)	The complex acts as a transcription factor

THE KIDNEY — THREE CHAINS

Ultrafiltration: the afferent arteriole is **wider** than the efferent → **high hydrostatic pressure** → small molecules forced through **fenestrations, the basement membrane and between podocytes**. The **basement membrane** stops plasma proteins.

Selective reabsorption (PCT): Na⁺ is **actively transported** out of the tubule cells into the blood → Na⁺ is low inside → Na⁺ diffuses in from the lumen through a **co-transport protein carrying glucose** → glucose enters the blood by facilitated diffusion. PCT cells have **microvilli** and **many mitochondria**.

Loop of Henle: the limbs flow in **opposite directions**; the ascending limb **actively transports Na⁺ and Cl⁻ out** — write “**ions**”, **not “sodium**” — and is **impermeable to water** → the medulla has a **low water potential** → water leaves the descending limb by osmosis → the effect is **multiplied** down the loop → water is drawn out of the **collecting duct**.

ADH: osmoreceptors in the hypothalamus detect a fall in blood water potential → ADH is **made in the hypothalamus and released from the posterior pituitary** → acts on the **collecting duct and DCT** (“loop of Henle” is ignored) → binds receptors → vesicles of **aquaporins** fuse with the membrane → more water reabsorbed → small volume of concentrated urine.

Desert mammals: longer loops and a thicker medulla, more pumps, a greater countercurrent effect, more aquaporins, more mitochondria.

NERVES, SYNAPSES AND THE BRAIN

RESTING AND ACTION POTENTIAL

Resting (−70 mV): the **Na⁺/K⁺ pump** moves **3 Na⁺ out and 2 K⁺ in** using ATP; voltage-gated Na⁺ channels are closed and the membrane is **more permeable to K⁺**, which diffuses out, leaving the inside negative.

Action potential: a stimulus opens some Na⁺ channels → **threshold (−55 mV)** → voltage-gated Na⁺ channels open (**positive feedback**) → **depolarisation to +40 mV** → Na⁺ channels close and **voltage-gated K⁺ channels open** → **repolarisation** → **hyperpolarisation** → the **refractory period** makes impulses discrete and one-way.

All-or-nothing: a stronger stimulus gives **more frequent** impulses, not bigger ones.

SYNAPTIC TRANSMISSION

Action potential reaches the presynaptic knob → **voltage-gated calcium ion channels open** and **Ca²⁺ diffuse in** → vesicles **move to and fuse with** the presynaptic membrane → neurotransmitter released by **exocytosis** → dif-

fuses across the **synaptic cleft** → binds **complementary receptors** → **ligand-gated Na⁺ channels open** → depolarisation → action potential if threshold is reached → neurotransmitter **broken down (acetylcholinesterase) or taken back up**.

"Sodium ion channel as a receptor" is ignored; "**sodium**" **unqualified is not accepted** — write **sodium ions**.

Myelination, three points: myelin is an insulator, so ions cannot cross where it is present; action potentials occur **only at the nodes of Ranvier**, where voltage-gated Na⁺ channels are concentrated; the impulse **jumps from node to node — saltatory conduction** — so it is faster. "A myelinated neurone has insulation" is ignored.

Drugs: nicotine mimics acetylcholine; **lidocaine and TTX block voltage-gated Na⁺ channels**; MDMA increases serotonin release; **SSRIs block reuptake proteins**; **L-DOPA crosses the blood-brain barrier and is converted to dopamine**, which cannot cross itself.

Parkinson's: death of dopamine-producing neurones in the **substantia nigra** → less dopamine binds receptors → fewer action potentials → less muscle control.

	How	Shows
CT	X-rays	Structure only
MRI	Magnetic field + radio waves	Soft-tissue structure
fMRI	Active areas get more oxyhaemoglobin , which absorbs radio waves differently	Activity in real time
PET	Radioactive tracer; active cells take up more	Metabolic activity

Rods: in the **dark** rhodopsin is intact, Na⁺ channels are **open**, the membrane is **depolarised**, inhibitory **glutamate is released** and the bipolar neurone is hyperpolarised — no impulse. In the **light** rhodopsin is **bleached to opsin and trans-retinal**, Na⁺ channels **close**, the membrane **hyperpolarises**, no glutamate is released and the bipolar neurone **depolarises** — impulse.

Pupil reflex: bright light → **circular muscles contract, radial relax** → pupil constricts. **Do not accept ciliary muscles.**

Habituation = a decreased response to a repeated, prolonged, **non-threatening** stimulus; fewer **Ca²⁺ channels** open → less neurotransmitter → less depolarisation.

PLANTS AND GENE EXPRESSION

	Pr (P660)	Pfr (P730)
Absorbs	Red , 660 nm	Far-red , 730 nm
Activity	Inactive	Active — promotes germination

THE RULE: THE LAST FLASH WINS

Last treatment red → Pfr high → germination. Last far-red

→ Pfr converted back to Pr → no germination.

The single most important rejection in this topic: "Pfr acts as a transcription factor" is not accepted. Pfr **activates** transcription factors; it is not one. Chain: Pfr enters the nucleus → activates a transcription factor → transcription of germination genes → gibberellin → amylase → starch broken down.

Auxin and phototropism: made in the shoot tip, **accumulates on the shaded side**, stimulates **cell elongation** there, so the shoot grows **towards the light** ("bends towards light" is not accepted). Mechanism: **H⁺ pumped into the wall** → **pH falls** → **hydrogen bonds between cellulose microfibrils break** → **water enters by osmosis** → **the cell elongates**.

Gibberellin: seed absorbs water → embryo releases gibberellin → acts via **transcription factors** in aleurone cells → the **amylase gene is switched on** → amylase hydrolyses **starch to maltose** → respired for ATP. "Produces energy" is ignored.

Transcription factors are proteins that bind the **promoter** so RNA polymerase can bind. **Epigenetics:** **DNA methylation** of cytosine silences genes; **histone acetylation** loosens DNA so transcription is possible. **The base sequence is unchanged.**

Alternative splicing: pre-mRNA → the **spliceosome removes introns and joins exons in different combinations** → different mRNAs → proteins with different primary structures. "DNA splicing" is not accepted.

RECOMBINANT DNA — THE 4-MARK CHAIN

1. Isolate the gene with a **restriction endonuclease** (or reverse transcriptase from mRNA → cDNA). **2.** Cut the **plasmid with the same enzyme** → complementary **sticky ends**. **3.** Join with **DNA ligase** (phosphodiester bonds). **4.** Insert into the host — bacteria by **heat shock or calcium chloride**; plants by *Agrobacterium* or a gene gun. **5.** Select transformed cells with marker genes.

"Restricting enzyme" is ignored; it is the **plasmid** that is inserted, not the gene directly.

Microarray route: extract mRNA → **reverse transcriptase** → **cDNA** → fluorescent label → hybridise → **fluorescing spots show active genes** → bioinformatics. Do not mix this route with the PCR route.

THE 20-MARK ARTICLE QUESTION

22% of the paper, in 6-10 parts tied to numbered paragraphs. **Budget 25 minutes.**

Sub-question type	What it is really asking
A Unit 5 chain in a new context (~50%)	Chemiosmosis, heart-rate control, the synapse, the kidney, sliding filament — dressed in the article's subject
A synoptic Units 1-4 chain (~30%)	Immune response, inflammation, protein structure, natural selection, osmosis, enzymes
Design a method	Antimicrobial test, habituation assay, toxicity assay
"State what is meant by..."	A term from the article
Small calculation or science in society	Ratio, % change, why a placebo is used

HOW TO PREPARE THE ARTICLE BEFOREHAND

1. Read it three times and **number every paragraph**. **2.** Write in the margin which topic each paragraph connects to. **3.** Write the mark-scheme chain next to every named process. **4.** Look up every unfamiliar term. **5.** For every experiment, ask: independent variable, controls, error bars. **6.** For every figure, practise one calculation.

The synoptic chains it reuses: **immune response** (antigen engulfed → presented to **T helper** cells → cytokines activate B cells → **plasma cells** release antibodies → memory cells); **inflammation (histamine** → vasodilation → **capillary walls** more permeable); **natural selection** (random mutation → selection pressure → **allele frequency increases over generations**); **protein solubility (hydrophilic R groups outside = soluble)**.

THE REJECTIONS BANK

Do not write	Write instead
"Signals" or "messages"	Impulses ; or waves of depolarisation for the SAN
"Produce energy"	Energy is released
"ATPase"	ATP synthase
"Sodium" unqualified	Sodium ions
ATP hydrolysis drives "the power stroke"	It returns the head; the power stroke is the release of ADP and P_i
"Pfr acts as a transcription factor"	Pfr activates transcription factors
Lactate "used in the Krebs cycle"	Converted to pyruvate in the liver first
"Control centre" unqualified	Name it: hypothalamus or medulla
"Ciliary muscles" in the pupil reflex	Circular and radial muscles

WALKING INTO THE EXAM

Protect 25 minutes for the article. Write chains **in the right order** and name the enzyme, the ion and the structure. Hedge every significance claim and add a methodology comment — it is often the cheapest mark on the page.